SIMATS ENGINEERING
Engineer to Excel

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Mobile Computing LAB

11.Design a wireframe for a mobile application that incorporates security measures using Figma

Aim:

To design a mobile application wireframe in Figma that incorporates security measures such as login authentication, password protection, and secure access.

Procedure:

1.Open Figma and create a new design file.

2.Create a mobile screen using a frame.

3.Design a login page with username and password fields.

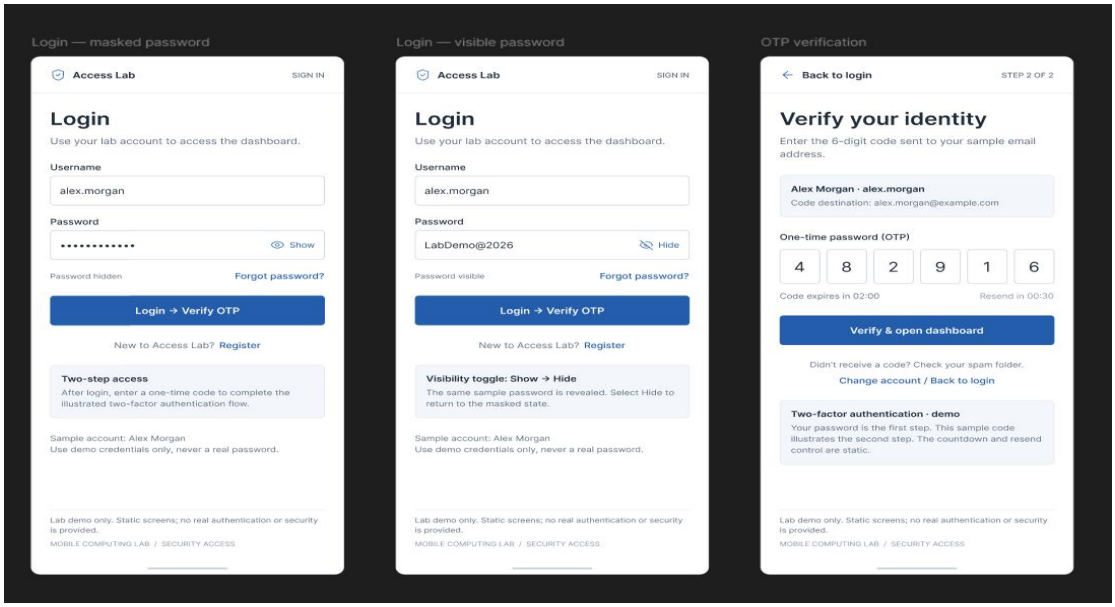
4.Add a login button, password visibility option, and forgot password link.

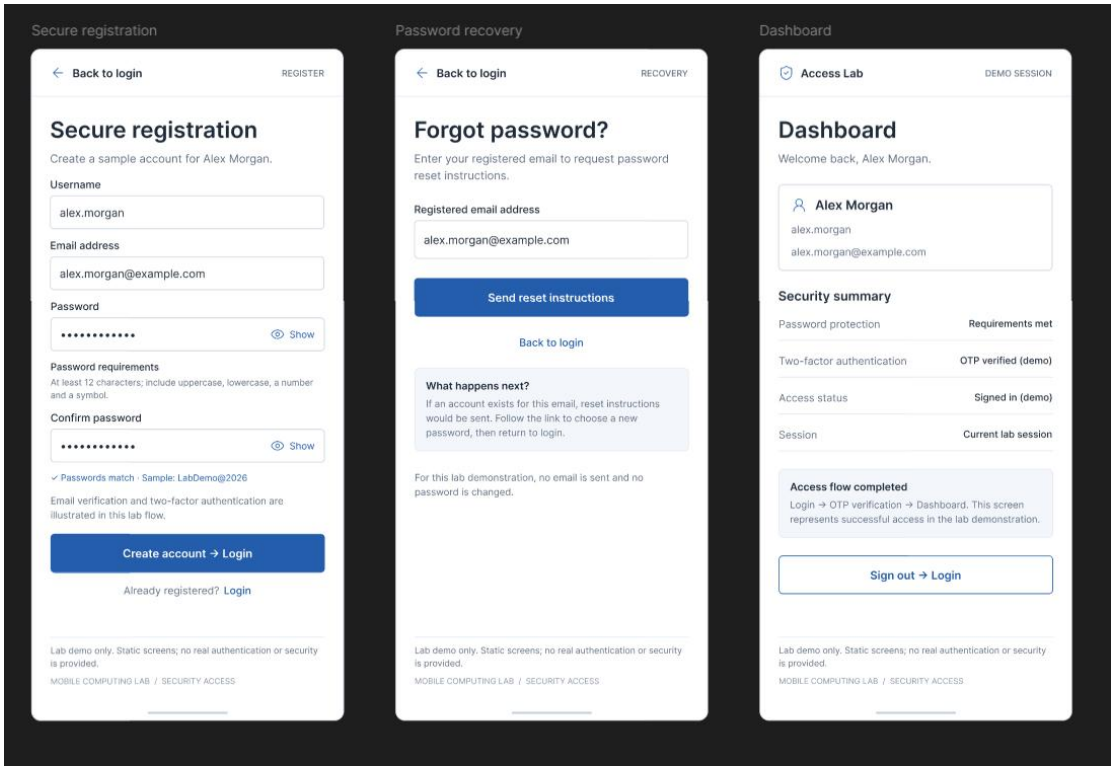
5.Design an OTP verification screen for two-factor authentication.

6.Connect the screens using prototype interactions.

7.Preview the design.

Design:





Result:

A secure mobile application wireframe was successfully designed using Figma.

12. Use Figma to Create an Interactive Diagram Illustrating the Evolution from 1G to 5G

Aim:

To create an interactive Figma diagram illustrating the evolution of mobile communication technologies from 1G to 5G.

Procedure:

1.Open Figma and create a new design file.

2.Create five frames representing 1G, 2G, 3G, 4G, and 5G.

3.Add the technology name, features, and data speed for each generation.

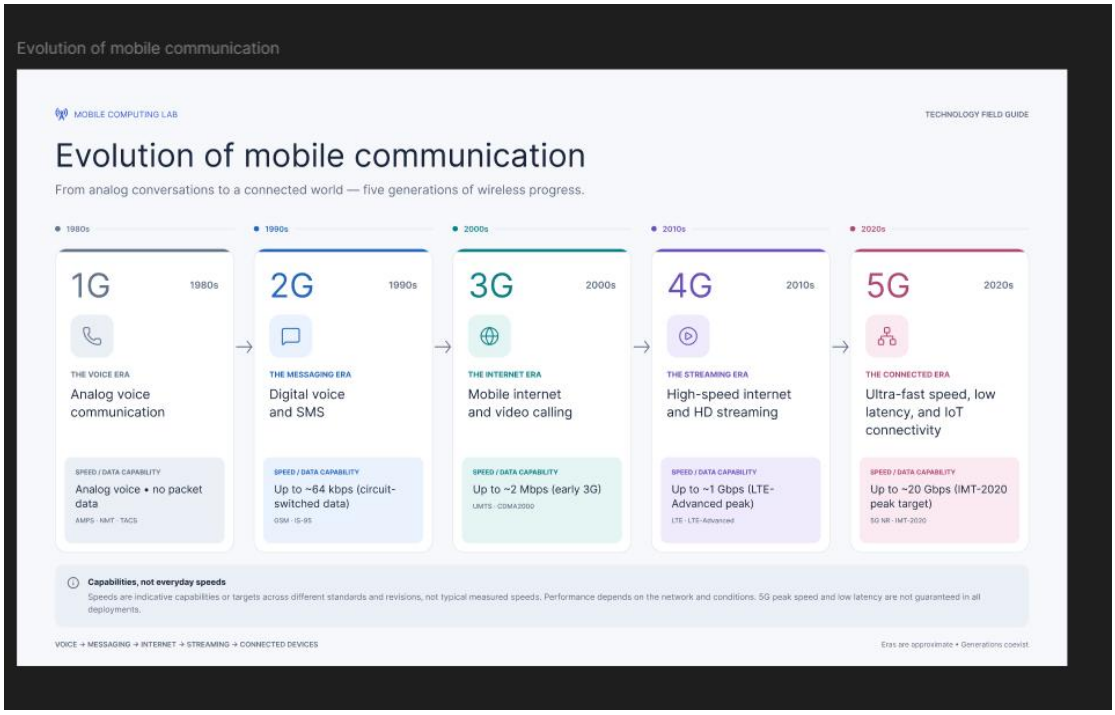
4.Arrange the frames in chronological order.

5.Add arrows connecting each generation.

6.Use prototype connections to make the diagram interactive.

7.Preview the diagram.

Design:



Result:

An interactive diagram illustrating the evolution of mobile communication from 1G to 5G was successfully created in Figma.

13. Design an Infographic in Figma Comparing Mobile Communication and Mobile Computing

Aim:

To design an infographic in Figma comparing mobile communication and mobile computing based on their features and applications.

Procedure:

1.Open Figma and create a new design file.

2.Create an infographic frame.

3.Divide the frame into two sections.

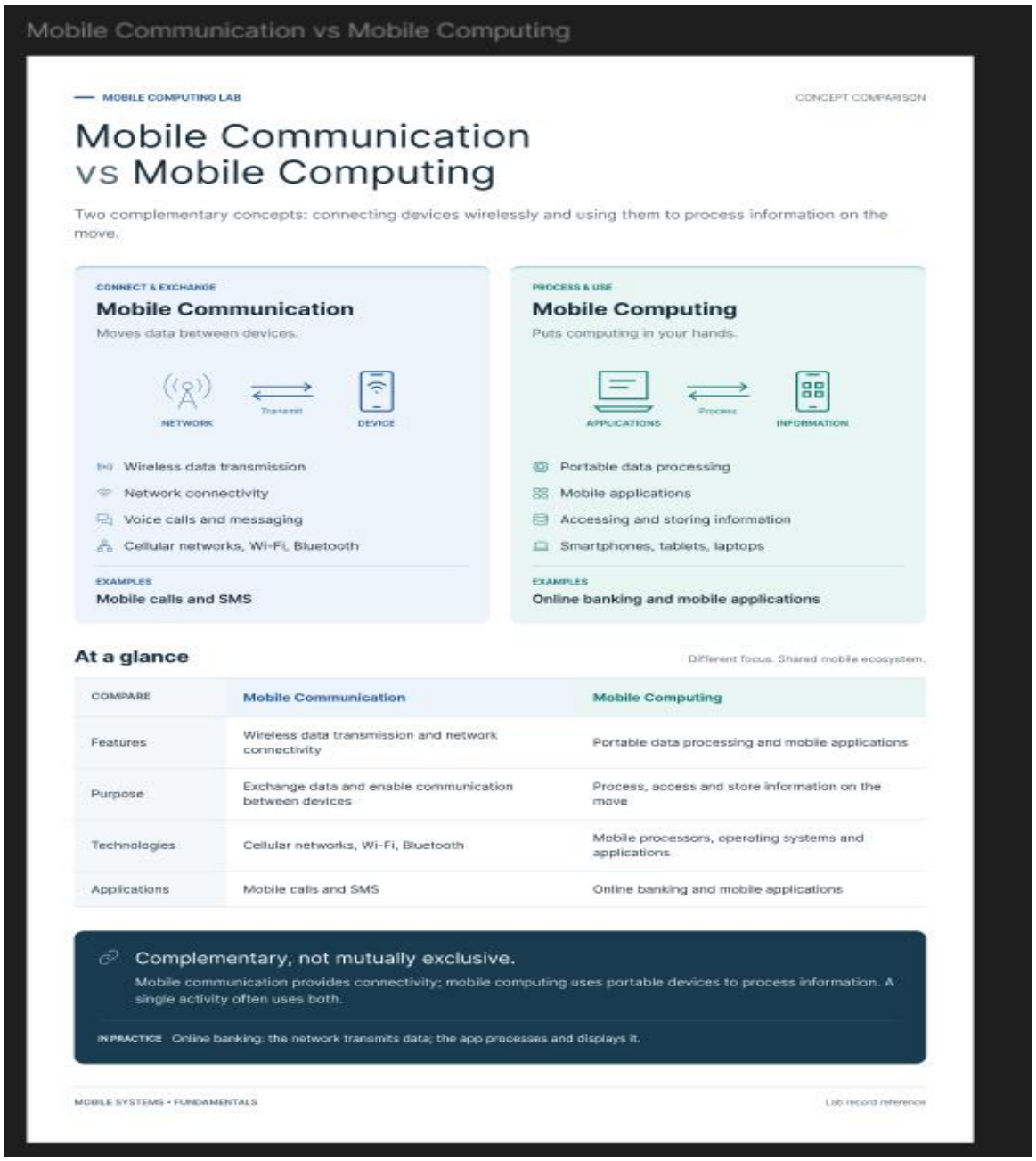
4.Add mobile communication features such as wireless transmission, networks, and connectivity.

5.Add mobile computing features such as data processing, applications, and portability.

6.Include suitable icons, colors, and illustrations.

7.Arrange the content neatly and preview the design.

Design:



Result:

An infographic comparing mobile communication and mobile computing was successfully designed using Figma.

14. Develop a Flowchart in Figma Explaining the Operation of GPRS

Aim:

To develop a flowchart in Figma explaining the operation of General Packet Radio Service (GPRS).

Procedure:

1.Open Figma and create a new design file.

2.Create a flowchart using standard symbols.

3.Add the following steps:

- Mobile Station (MS)
- Base Station Subsystem (BSS)
- Serving GPRS Support Node (SGSN)
- Gateway GPRS Support Node (GGSN)
- Internet

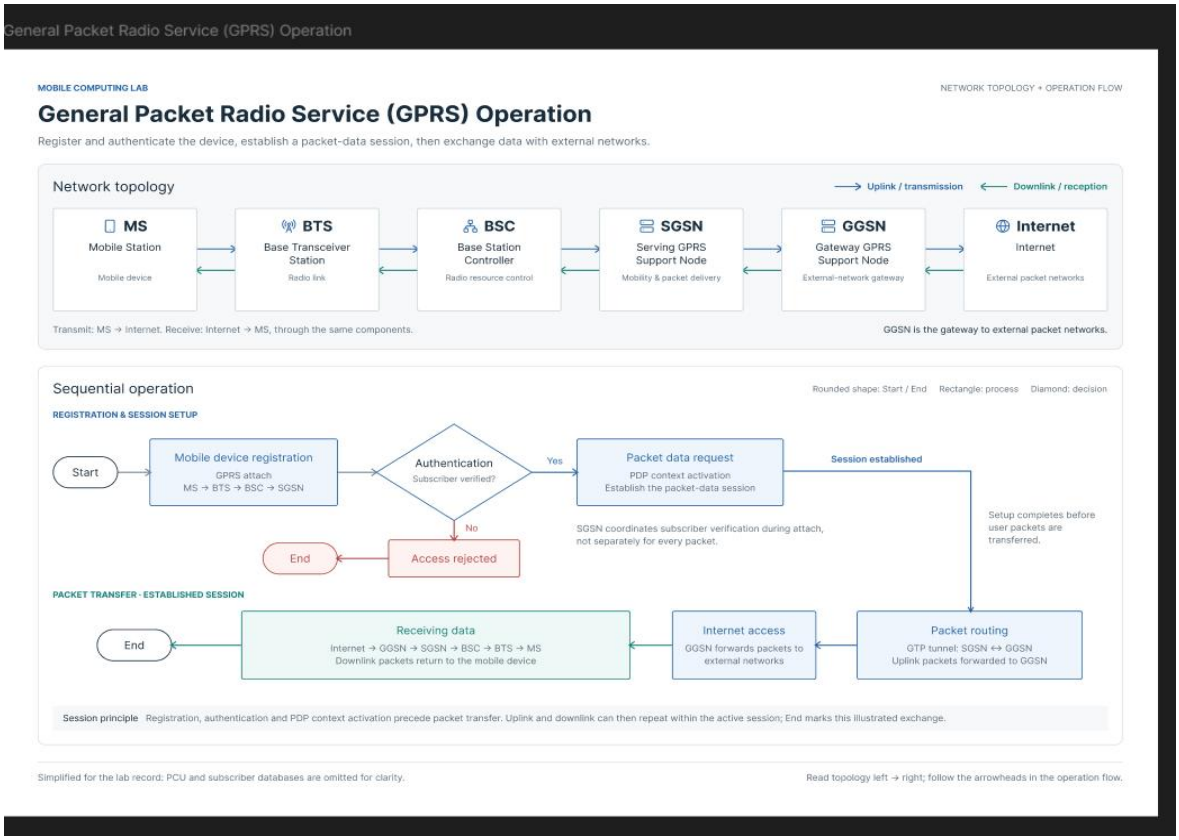
4.Connect the components using arrows.

5.Add labels explaining packet transmission.

6.Arrange the flowchart clearly.

7.Preview the final design.

Design:



Result:

A flowchart explaining the operation of GPRS packet transmission was successfully developed using Figma.

15. Develop a Wireframe for a Mobile Application Addressing Challenges Related to Ad Hoc Wireless Networks Using Figma

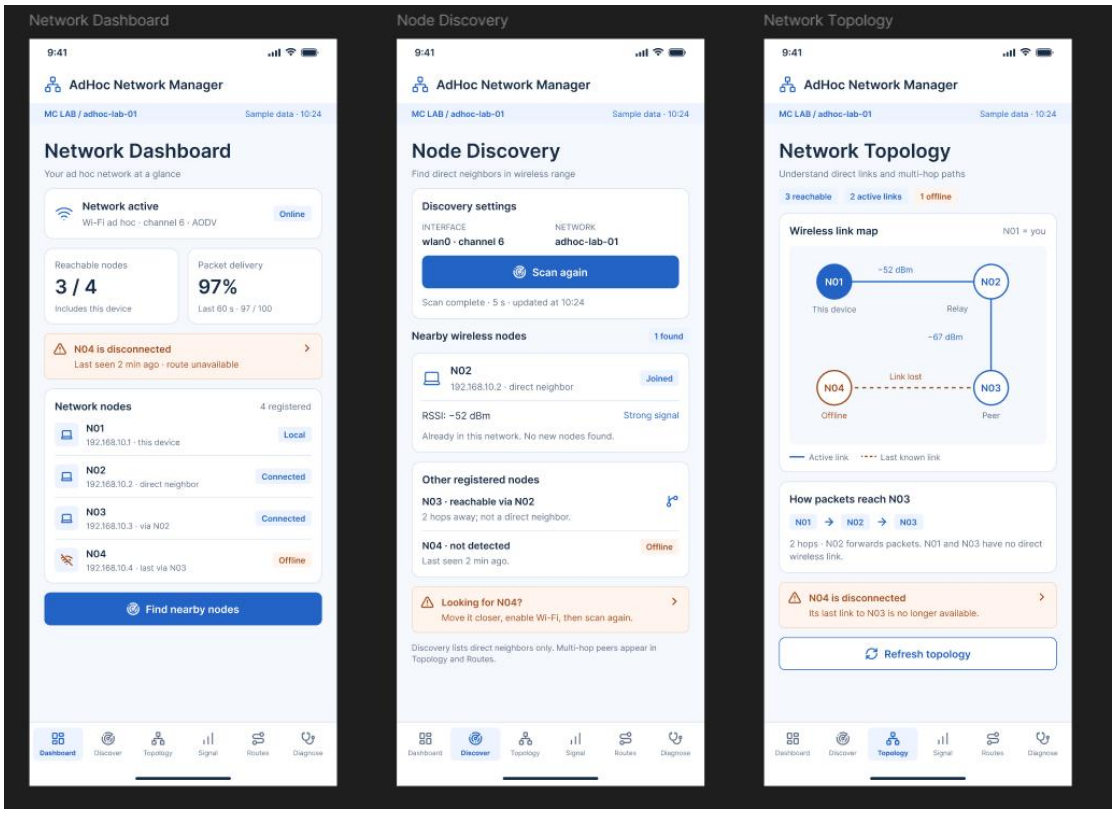
Aim:

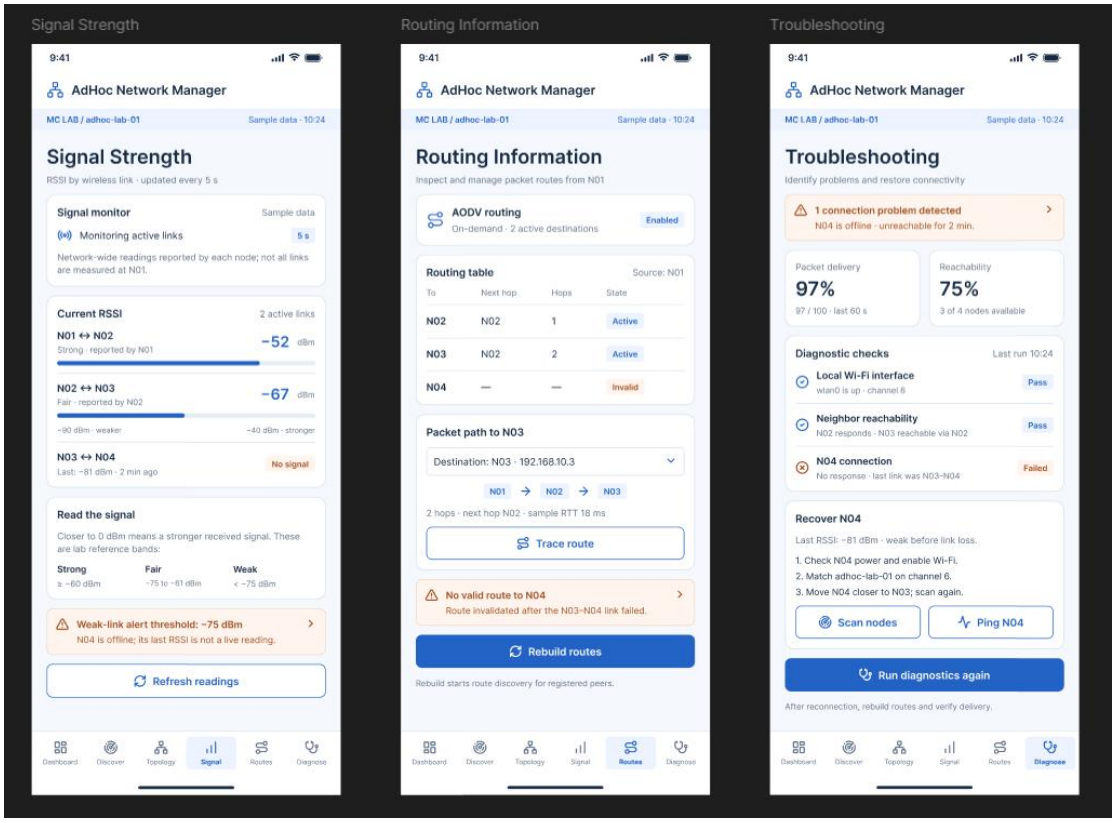
To design a mobile application wireframe in Figma that addresses challenges in ad hoc wireless networks, such as node connectivity, routing, and network reliability.

Procedure:

1. Open Figma and create a new design file.
2. Create a mobile application frame.
3. Design a home screen displaying network status.
4. Add screens for node discovery, routing information, and connection status.
5. Include features for monitoring signal strength and identifying disconnected nodes.
6. Add buttons for reconnecting nodes and troubleshooting network issues.
7. Connect the screens using prototype interactions.
8. Preview the wireframe.

Design:





Result:

A mobile application wireframe addressing ad hoc wireless network challenges was successfully designed using Figma.

16. Design a Dashboard Displaying Network Performance Metrics

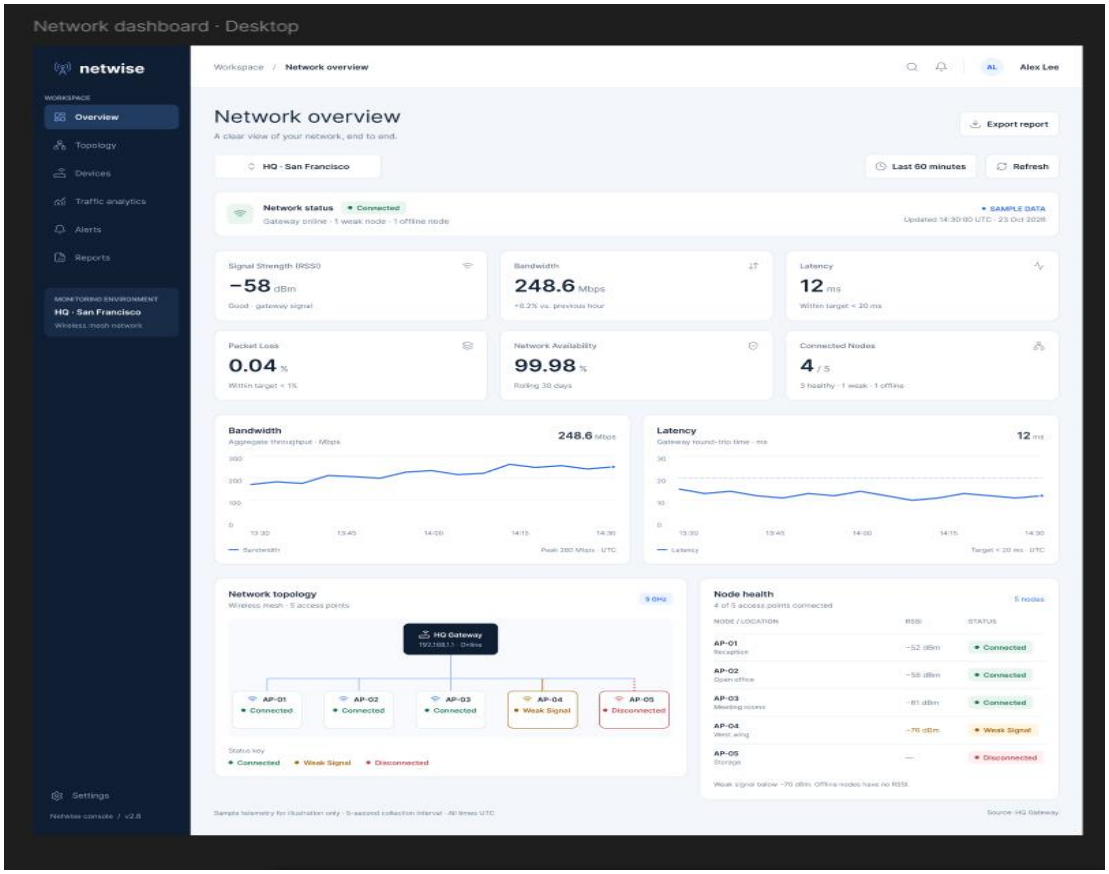
Aim:

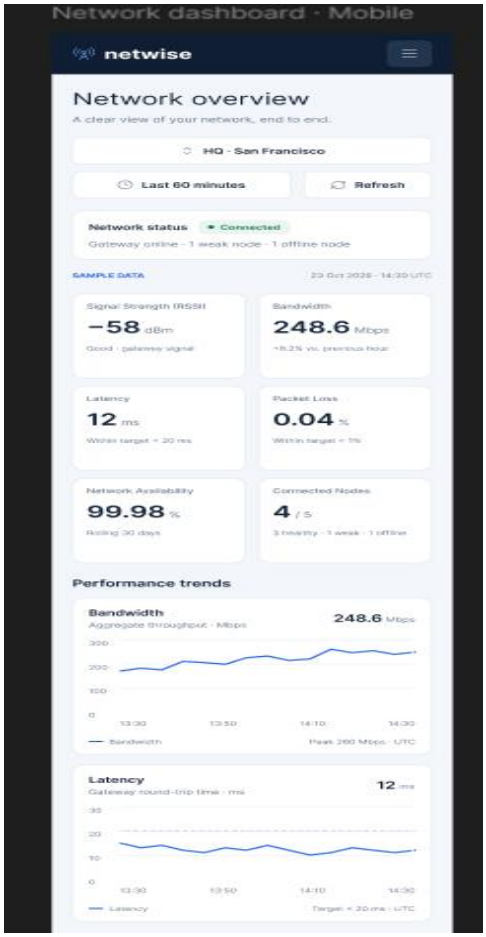
To design a dashboard displaying network performance metrics such as signal strength, bandwidth, latency, and packet loss.

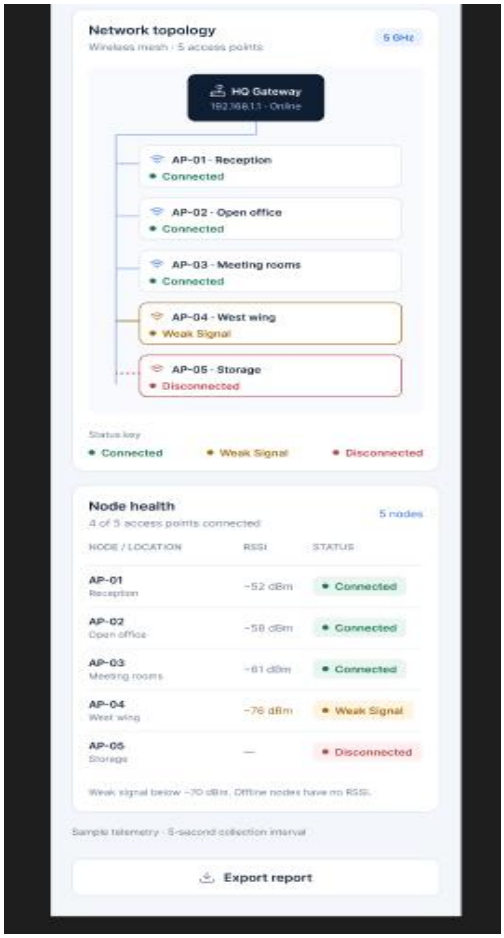
Procedure:

1. Open Figma and create a new design file.
2. Create a dashboard frame.
3. Add metric cards for signal strength, bandwidth, latency, and packet loss.
4. Include graphs to represent network performance.
5. Add network status indicators.
6. Arrange the components using proper alignment and spacing.
7. Preview the dashboard.

Design:


Network dashboard · Desktop
netwise
Workspace / Network overview
Alex Lee
Overview
Topology
Devices
Traffic analytics
Alerts
Reports
HQ · San Francisco
Wireless mesh network
Settings
Network overview
A clear view of your network, end to end.
Export report
HQ · San Francisco
Last 60 minutes
Refresh
Network status
Connected
Gateway online · 1 weak node · 1 offline node
SAMPLE DATA
Signal Strength (RSSI)
−58 dBm
Good · gateway signal
Bandwidth
248.6 Mbps
+8.2% vs. previous hour
Latency
12 ms
Within target < 20 ms
Packet Loss
0.04 %
Within target < 1%
Network Availability
99.98 %
Rolling 30 days
Connected Nodes
4 / 5
3 healthy · 1 weak · 1 offline
Bandwidth
Aggregate throughput · Mbps
248.6 Mbps
Latency
Gateway round-trip time · ms
12 ms
Network topology
Wireless mesh · 5 access points
5 GHz
HQ Gateway
192.168.1.1 · Online
AP-01
Connected
AP-02
Connected
AP-03
Connected
AP-04
Weak Signal
AP-05
Disconnected
Status key
Connected
Weak Signal
Disconnected
Node health
4 of 5 access points connected
5 nodes
NODE / LOCATION
RSSI
STATUS
AP-01
Reception
−52 dBm
Connected
AP-02
Open office
−58 dBm
Connected
AP-03
Meeting rooms
−61 dBm
Connected
AP-04
West wing
−76 dBm
Weak Signal
AP-05
Storage
—
Disconnected
Weak signal below −70 dBm. Offline nodes have no RSSI.
Sample telemetry for illustration only · 5-second collection interval · All times UTC
Source: HQ Gateway
Settings
Netwise console / v2.8


Network dashboard · Mobile
netwise
Network overview
A clear view of your network, end to end.
HQ · San Francisco
Last 60 minutes
Refresh
Network status
Connected
Gateway online · 1 weak node · 1 offline node
SAMPLE DATA
Signal Strength (RSSI)
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Rolling 30 days
Connected Nodes
4 / 5
Performance trends
Bandwidth
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Network topology
Wireless mesh · 5 access points
5 GHz
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AP-01 · Reception
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AP-02 · Open office
Connected
AP-03 · Meeting rooms
Connected
AP-04 · West wing
Weak Signal
AP-05 · Storage
Disconnected
Status key
Connected
Weak Signal
Disconnected
Node health
4 of 5 access points connected
5 nodes
NODE / LOCATION
RSSI
STATUS
AP-01
Reception
−52 dBm
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Open office
−58 dBm
Connected
AP-03
Meeting rooms
−61 dBm
Connected
AP-04
West wing
−76 dBm
Weak Signal
AP-05
Storage
—
Disconnected
Weak signal below −70 dBm. Offline nodes have no RSSI.
Sample telemetry · 5-second collection interval
Export report

Result:

A network performance dashboard displaying important network metrics was successfully designed using Figma.

17. Create an Error Alert UI for Connection Drops

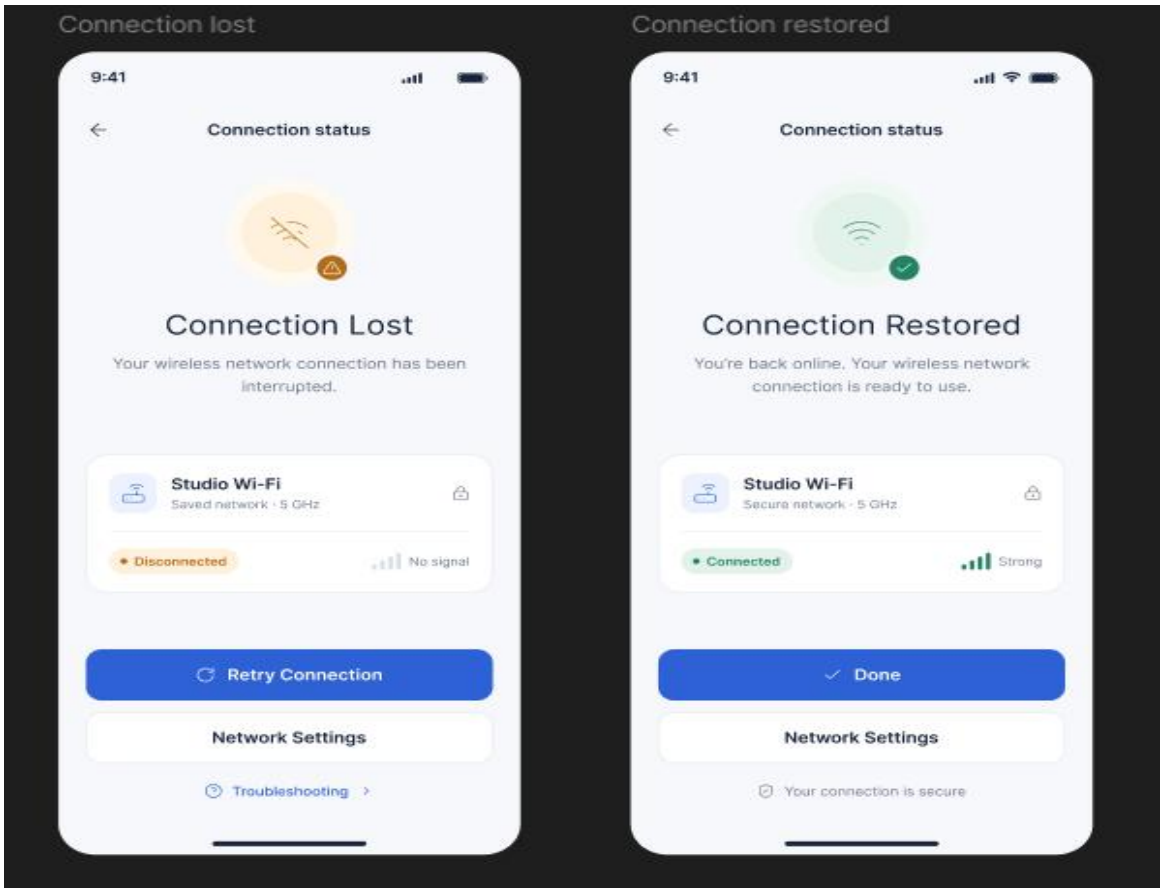
Aim:

To create an error alert user interface in Figma for notifying users about network connection drops.

Procedure:

1. Open Figma and create a new design file.
2. Create a mobile application frame.
3. Design an error alert box.
4. Add a warning icon and the message "Connection Lost".
5. Include buttons such as "Retry" and "Network Settings".
6. Use suitable colors and typography to highlight the error.
7. Create prototype interactions for the buttons.
8. Preview the alert UI.

Design:



Result:

An error alert UI for connection drops was successfully created using Figma.

18. Simulate a Basic Ad Hoc Network Using a Networking Tool

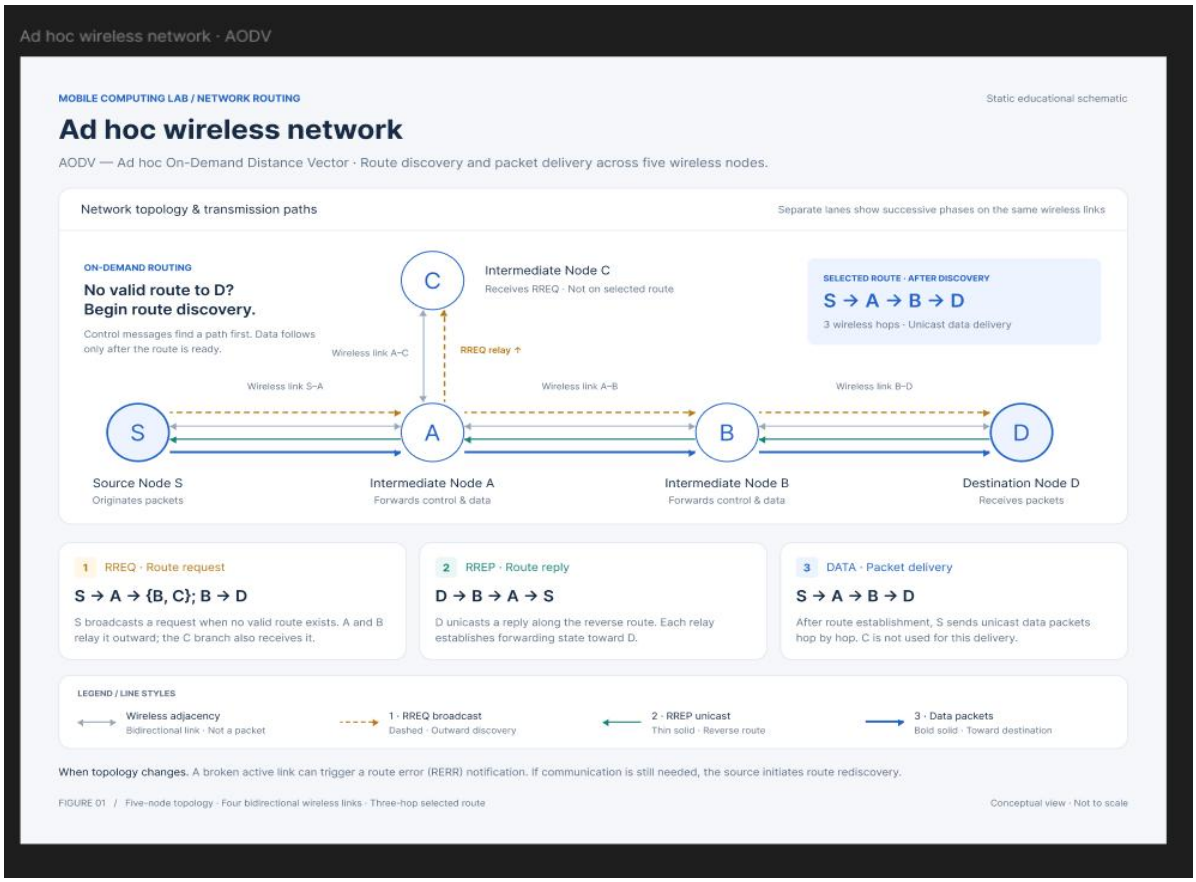
Aim:

To simulate a basic ad hoc wireless network using a networking simulation tool.

Procedure:

1. Open a networking simulation tool such as NS-2 or NS-3.
2. Create multiple wireless nodes.
3. Configure the wireless channel and network interface.
4. Set up an ad hoc routing protocol such as AODV.
5. Establish communication between the nodes.
6. Generate traffic between source and destination nodes.
7. Run the simulation and observe packet transmission.

Design:



Result:

A basic ad hoc wireless network was successfully simulated using a networking tool.

19. Analyze Signal Strength Between Nodes in a Simulated Environment

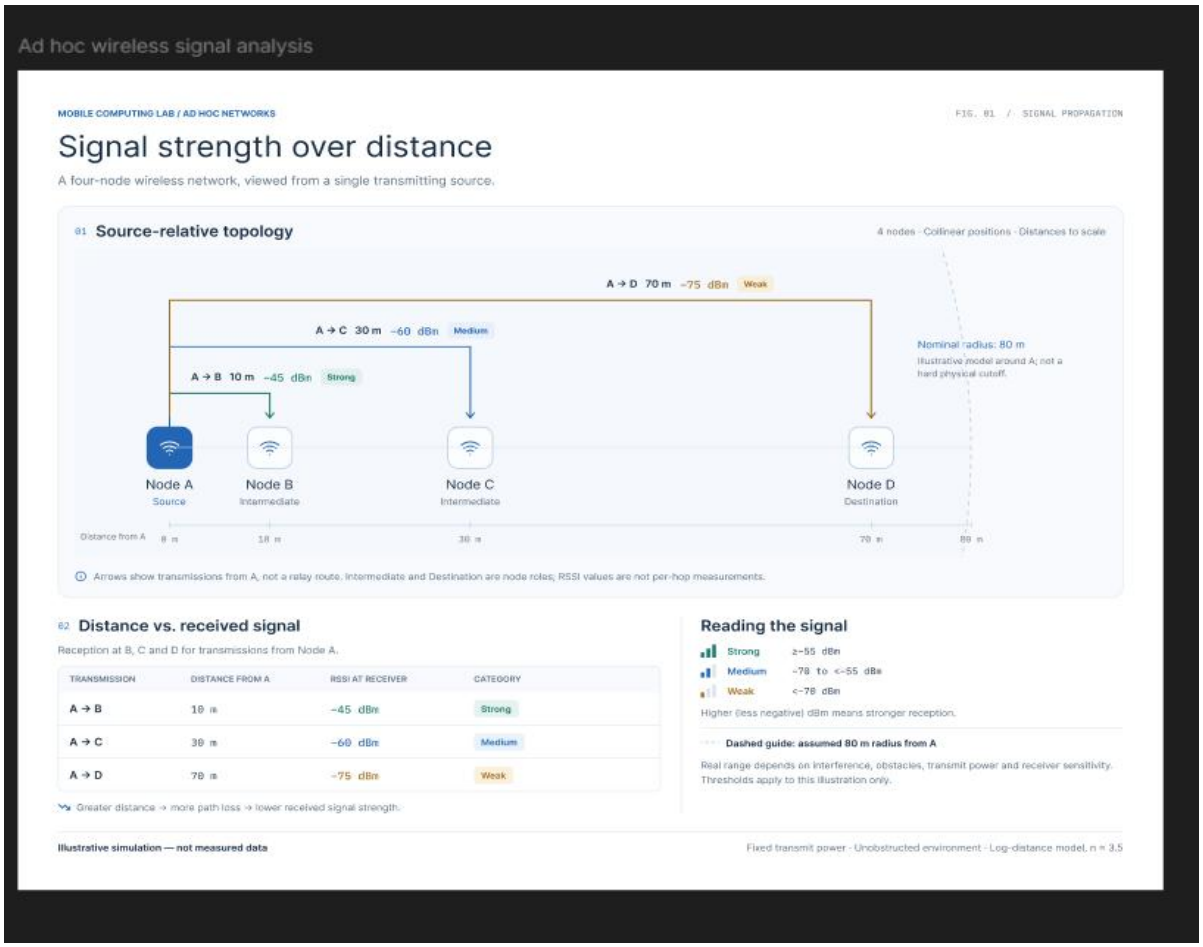
Aim:

To analyze the signal strength between wireless nodes in a simulated ad hoc network environment.

Procedure:

1. Open a networking simulation tool.
2. Create multiple wireless nodes.
3. Configure transmission power and communication range.
4. Establish communication between the nodes.
5. Vary the distance between the nodes.
6. Observe the received signal strength (RSSI).
7. Record and compare the signal strength values.

Design:



Result:

The signal strength between wireless nodes was successfully analyzed in a simulated environment.

20. Develop a Troubleshooting Flow for Common Issues in Ad Hoc Networks

Aim:

To develop a troubleshooting flowchart for identifying and resolving common issues in ad hoc wireless networks.

Procedure:

1.Open Figma and create a new design file.

2.Create a flowchart frame.

3.Add the starting point as "Network Problem Detected".

4.Add decision blocks for checking node connectivity, signal strength, routing, and packet loss.

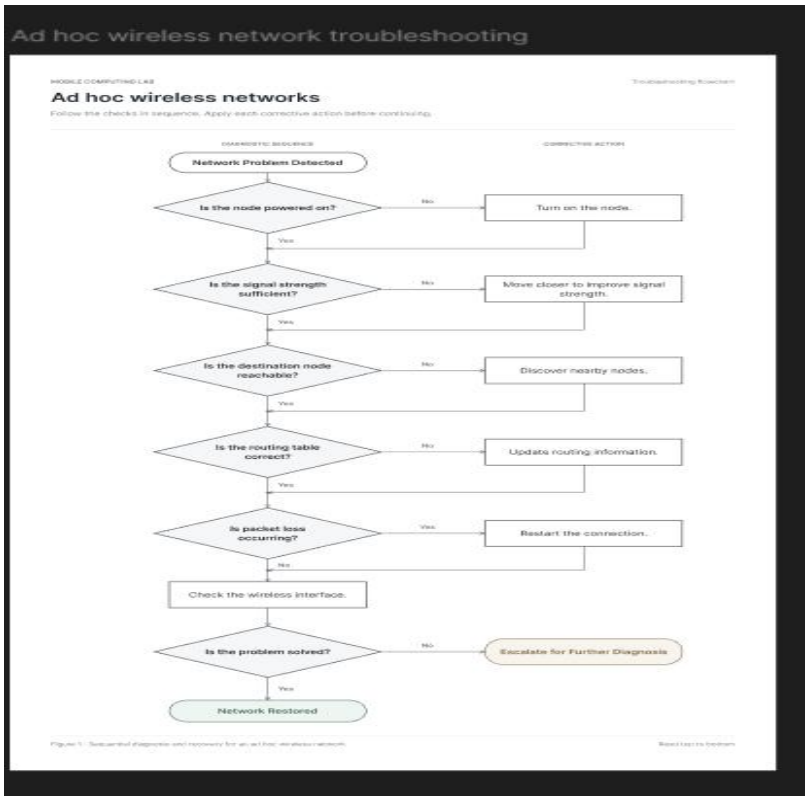
5.Add corrective actions such as reconnecting nodes and updating routing information.

6.Connect the blocks using arrows.

7.Add the final step as "Network Restored".

8.Preview the flowchart.

Design:



Result:

A troubleshooting flowchart for resolving common ad hoc wireless network issues was successfully developed using Figma.